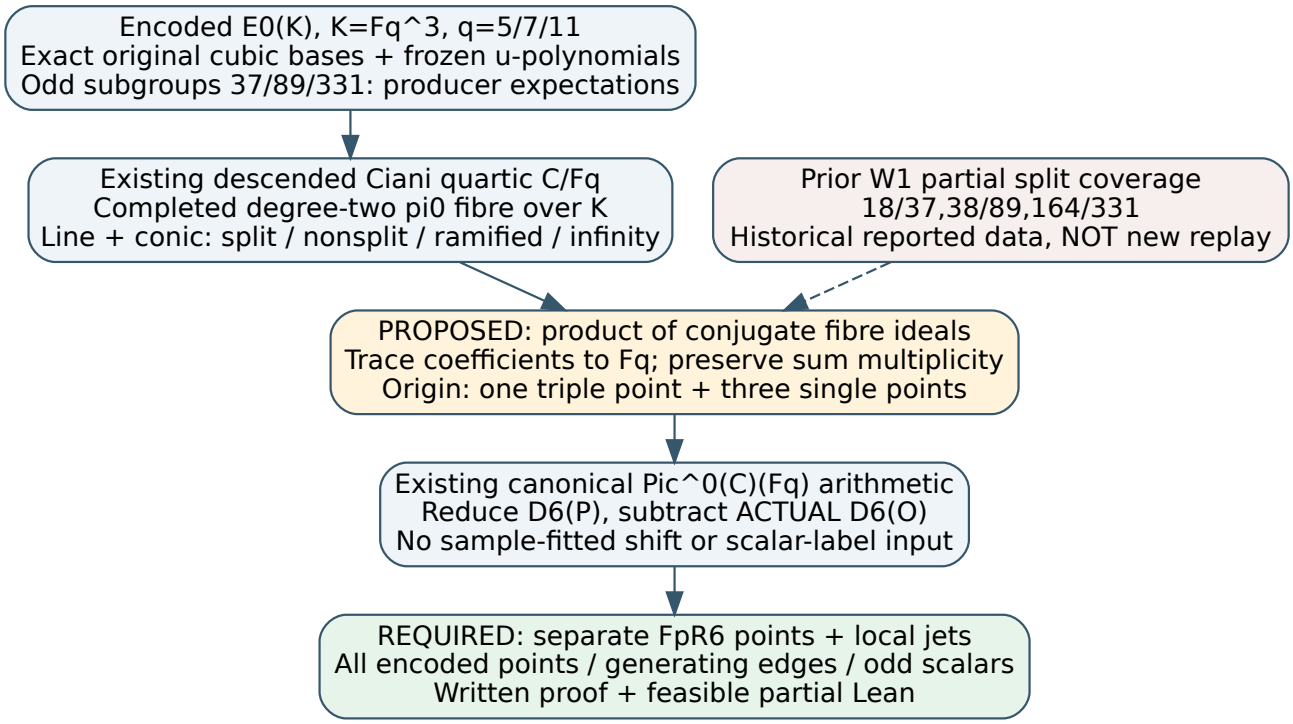
Complete encoded genus-3 descent

Source review and frozen proposal · 2026-11-13 · unsigned · no new native run



The known GHS/Ciani construction and canonical plane-quartic Picard arithmetic already exist. The next component completes their encoded-point transfer, including nonsplit, ramified, infinity and overlapping Frobenius fibres, with an actual origin shift. No new native arithmetic run, novelty or performance claim is made. Scientific standing is unsigned. [pcc:unverified]

The initial public panel is the first split case at q=5,7,11 in the original build record. It reports source orders148,356,1324 and odd primes37,89,331. These are producer expectations to independently recompute. Finalized candidate/EC1 identities remain null pending the frozen generator rule. Exact original cubic bases and complete records are in cases.json. [pcc:unverified]

Sources, controls and exact remaining work

The nineteen-source catalog records exact repository hashes, commits, literature and access scope. The program already separates direct GHS/Ciani descent from the Smith route. The actual canonical Picard source supplies degree-six reduction and full group operations. Its historical V4 record covers18/37,38/89,164/331 split inputs and calibrates an origin shift from samples. The old quartic adapter's elliptic operations consult a scalar table. Those records are prior producer evidence, not new acceptance or a new independent oracle. [pcc:unverified]

Joux–Vitse's author PDF, section2.1 and4.1, was downloaded and read after web-fetch failures. It supplies the known norm-conorm setting, subgroup condition and nonhyperelliptic genus-three family, while recording a decomposition limitation. Chao/Momose ECC2007, slides6–7,17,25–26, supplies the covering context and hypotheses. These are sources for prior context; the proposed adapter needs its own written and native checks. [pcc:unverified]

Six actual KB commands return127 with no backend executable; complete commands/output remain. The atlas remains vault-locked. Neither access gap establishes novelty or absence. Two tracked sparse source blobs were read through git show after the sparse helper returned2 for a missing profile marker. External PR4015 was read as separate cross-characteristic work and was not executed or adopted here. [pcc:unverified]

Proposed method: represent a completed degree-two fibre by a line and conic over $K=Fq^3$. Multiply its three conjugate ideal sheaves, homogenize their eight products to degree six, and descend the span by coefficient traces. Use the existing quartic reducer, then subtract the actual completed origin fibre. A product preserves summed local multiplicity. At the origin one shared point occurs three times and three other points occur once; a reduced union has degree4 rather than6. The independent oracle uses FpR6 point supports and local jets. This derivation has not yet received new native or Lean acceptance. [pcc:unverified]

The source review found two pre-run hazards. Historical cubic bases can differ from the default current constructor, so coefficient arrays must never be reinterpreted. The old square-root shortcut is documented only for $q^3=3$ modulo4 and cannot serve q=5. Reuse the existing FpR arithmetic and general square root. The quartic-infinity Weierstrass model also has two different x_unshifted=0 branches, one finite and one infinite. [pcc:unverified]

The protocol requires every encoded source point, independent counts, coordinate roundtrips, complete divisor schemes, every edge under an independently checked generating set, all odd-subgroup scalars and exact image order, plus negative and coherent positive controls. It preserves794 current prior-full inputs and the WHOLE20796-byte six-job consumer workflow. Only named visibility/accessor additions are planned in existing modules. Existing arithmetic/rank/verifier criteria remain unchanged. [pcc:unverified]

The written derivation treats actual covers, the origin, Cartier-ideal products and descent, and V4 pullback/norm composites[2]. Cross terms use character separation, not coprime conjugate elliptic orders. The resulting odd-subgroup injectivity must be identified with the actual geometric objects. Feasible cleared polynomial and conditional group Lean lemmas are owed. Full Picard/sheaf/descent/reducer formalization and an arbitrary-class norm remain distinct gaps. [pcc:unverified]

Next publish the protocol before code, implement the adapter, and retain bounded native correctness/proof attempts in full. A destination IC solve and admitted matched cold costs remain later tasks; no paid launch is admitted. Existing C0/T20 canonical graphs, fp-chain figures and consumer fixture routes remain current: this proposal changes no accepted map or cost edge. Editable DOT/SVG, recorded metadata, Typst and the inspected PDF accompany this report. [pcc:unverified]